

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 5: Construct interoperable web services by leveraging JAX-RPC, WSDL, SOAP, and XML Schema for seamless data exchange across distributed systems. (L4, L5)

Assessment Tool : Industry Problem Task

Weightage: 40%

QUESTIONS:

Total Marks: 30

1. A hospital management system requires a **SOAP-based web service** to manage patient information such as registration, appointment scheduling, and medical records. The system must ensure secure, structured, and interoperable communication between different hospital departments.

Design a **SOAP-based web service** for the hospital system. Explain the structure of SOAP messages (Envelope, Header, Body), define the operations involved (e.g., add patient, get records), and illustrate how data is exchanged between client and server. Justify the use of SOAP in this scenario considering reliability and security.

2. An enterprise application needs to expose its web services so that external systems can understand how to interact with them. The organization plans to use **WSDL (Web Services Description Language)** to describe the service.

Design the **WSDL structure** for a web service that provides operations such as retrieving and updating data. Explain the key components of WSDL (types, message, portType, binding, service), and illustrate how they define the service interface and communication details. Provide a clear structure and justify its importance in service integration.

3. A company is experiencing issues in its SOAP-based communication system where requests are not processed correctly, and responses are either delayed or incorrect.

Analyze the SOAP communication process and identify possible issues such as message formatting errors, incorrect namespaces, or endpoint problems. Propose a debugging approach to resolve these issues, including validation of SOAP messages and error handling techniques. Explain how the corrected system ensures reliable communication.

Code of Conduct:

I Deepak V-192411021 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:**RUBRICS FOR CALCULATION:**

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

1. SOAP-BASED WEB SERVICE FOR HOSPITAL MANAGEMENT SYSTEM

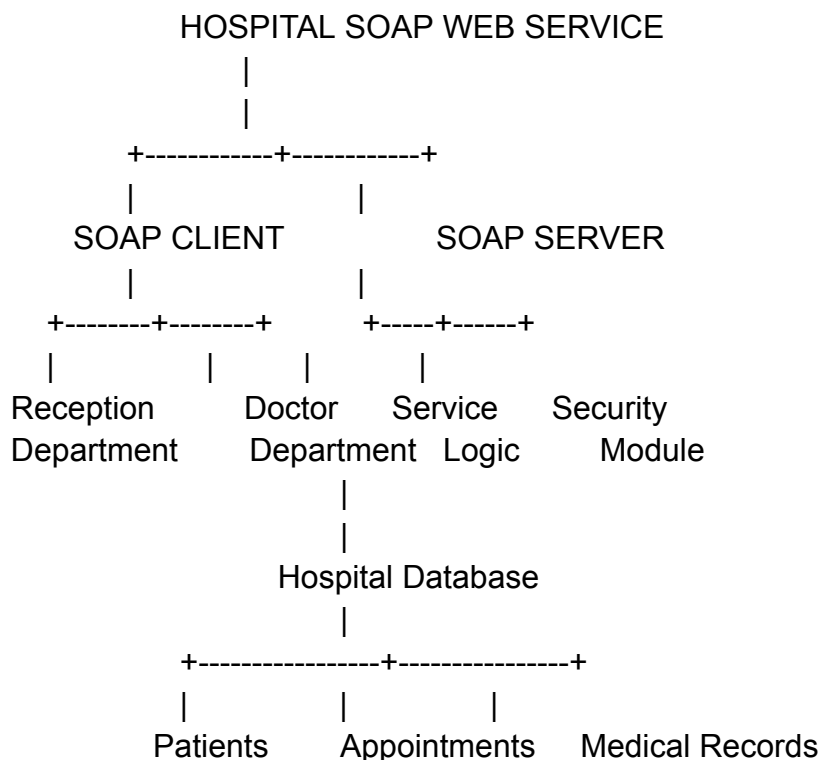
INTRODUCTION

A hospital management system can use a SOAP-based web service to allow different departments such as reception, doctors, pharmacy, laboratory, billing, and medical records to communicate securely. SOAP provides a standard XML-based communication mechanism that allows systems developed using different programming languages and platforms to exchange information.

OBJECTIVES OF THE SYSTEM

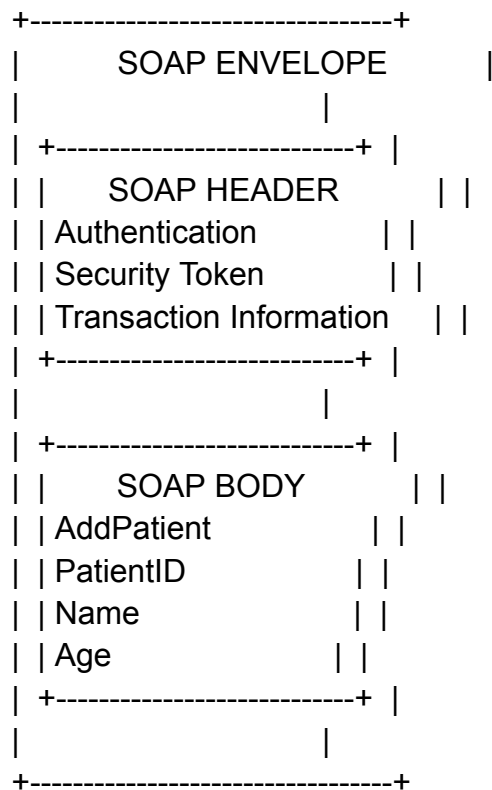
- Register and maintain patient information.
- Schedule and manage patient appointments.
- Retrieve and update medical records.
- Provide secure communication between hospital departments.
- Maintain reliable and standardized data exchange.

BASIC ARCHITECTURE



SOAP MESSAGE STRUCTURE

A SOAP message consists mainly of an Envelope, Header, and Body. The Header is optional, while the Envelope and Body are required.



COMPONENTS OF SOAP MESSAGE

Component	Purpose
Envelope	Identifies the message as a SOAP message and contains Header and Body
Header	Contains optional security, authentication, and transaction information
Body	Contains the actual request or response
Fault	Contains information about errors that occurred during processing

OPERATIONS PROVIDED BY THE SERVICE

1. AddPatient
Registers a new patient in the hospital database.
2. GetPatient
Retrieves the basic details of a registered patient.
3. ScheduleAppointment
Creates an appointment between a patient and doctor.

4. GetMedicalRecords

Retrieves the medical history and records of a patient.

5. UpdateMedicalRecord

Allows authorized medical staff to update patient records.

OPERATION TABLE

Operation	Input	Output	Purpose
AddPatient	Patient details	Patient ID / Status	Register patient
GetPatient	Patient ID	Patient details	Retrieve patient information
ScheduleAppointment	Patient ID, Doctor ID, Date	Appointment ID	Schedule appointment
GetMedicalRecords	Patient ID	Medical records	Retrieve medical history
UpdateMedicalRecord	Patient ID, Record	Status	Update medical information

SOAP REQUEST EXAMPLE

```
<soap:Envelope
  xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">

  <soap:Header>
    <Authentication>
      <Username>hospitalUser</Username>
      <Token>ABC123</Token>
    </Authentication>
  </soap:Header>

  <soap:Body>
    <AddPatient>
      <PatientID>P101</PatientID>
      <Name>Rahul</Name>
      <Age>35</Age>
      <Gender>Male</Gender>
      <Contact>9876543210</Contact>
    </AddPatient>
  </soap:Body>

</soap:Envelope>
```

SOAP RESPONSE EXAMPLE

```

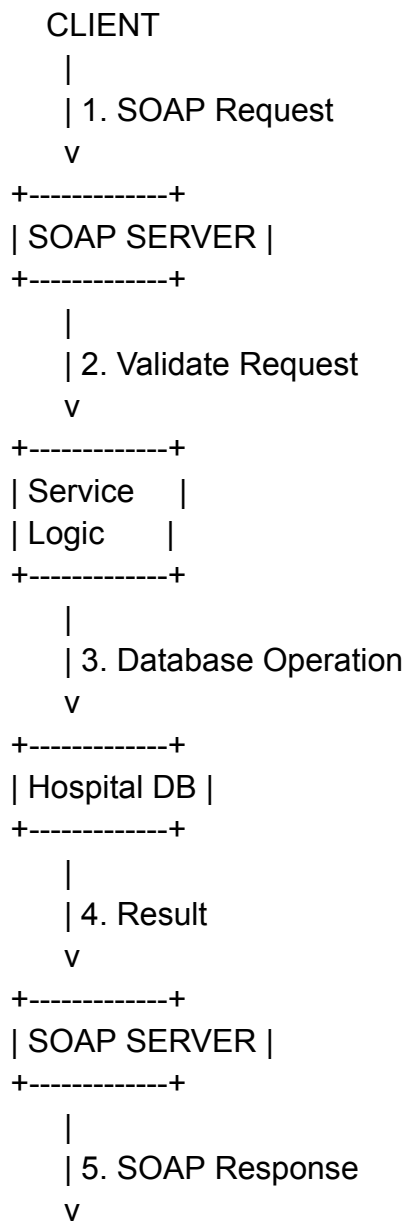
<soap:Envelope
  xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">

  <soap:Body>
    <AddPatientResponse>
      <PatientID>P101</PatientID>
      <Status>Success</Status>
      <Message>Patient added successfully</Message>
    </AddPatientResponse>
  </soap:Body>

</soap:Envelope>

```

DATA EXCHANGE PROCESS



CLIENT

STEP-BY-STEP COMMUNICATION

- The hospital client creates a SOAP request containing the required patient information.
- The SOAP request is placed inside the SOAP Envelope and Body.
- Security and authentication information can be placed in the SOAP Header.
- The request is sent to the hospital web-service endpoint.
- The SOAP server receives and validates the request.
- The server executes the required operation.
- The server communicates with the hospital database.
- The result is placed inside a SOAP response.
- The response is sent back to the client.
- If processing fails, a SOAP Fault is returned.

SECURITY FEATURES

- WS-Security can be used to provide message-level security.
- Authentication can verify the identity of the client.
- Encryption protects sensitive patient information.
- Digital signatures can ensure message integrity.
- HTTPS can protect communication during network transmission.
- Authorization can ensure that only permitted staff access medical records.

WHY SOAP IS SUITABLE FOR A HOSPITAL SYSTEM

Requirement	SOAP Advantage
Security	Supports WS-Security, authentication, encryption, and signatures
Reliability	Supports standardized error handling using SOAP Fault
Interoperability	Works across different platforms and programming languages
Structured data	Uses XML-based structured messages
Transactions	Can support enterprise-level transaction mechanisms
Standardization	Uses well-defined web-service standards

CONCLUSION

SOAP is suitable for a hospital management system because healthcare information is sensitive and requires secure, structured, and reliable communication. SOAP provides XML-based messaging, security mechanisms, standardized error handling, and interoperability between different hospital systems and departments.

2. WSDL STRUCTURE FOR AN ENTERPRISE WEB SERVICE

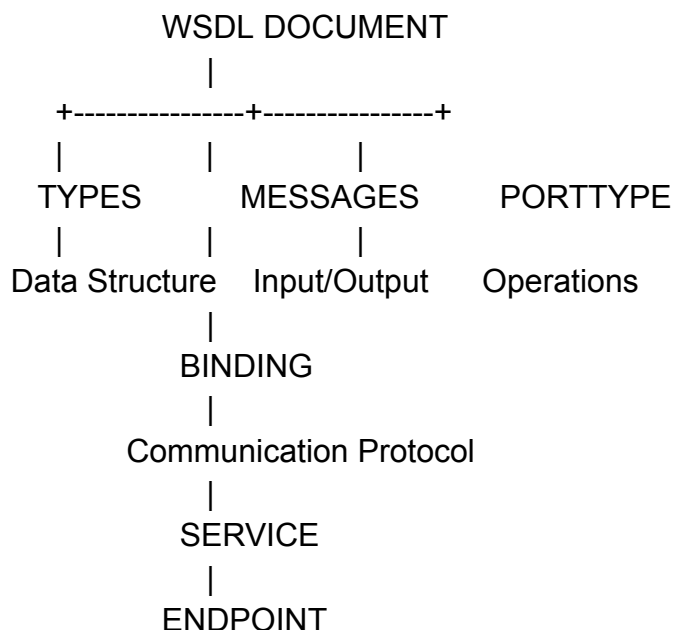
INTRODUCTION

WSDL (Web Services Description Language) is an XML-based language used to describe a web service and explain how clients can communicate with it. It acts as a formal contract between the service provider and service consumer by defining operations, messages, data types, communication protocols, and service endpoints.

PURPOSE OF WSDL

- Describes the operations provided by a web service.
- Defines the input and output messages.
- Specifies the communication protocol and message format.
- Provides the network address of the service.
- Helps different applications understand and consume the service automatically.

WSDL ARCHITECTURE



MAIN COMPONENTS OF WSDL

Component	Description
Types	Defines data types used by the service

Message	Defines the data exchanged between client and server
PortType	Defines operations and their input/output messages
Binding	Defines communication protocol and message format
Service	Defines the actual service endpoint

1. TYPES

The Types section defines the data structures used by the service. XML Schema (XSD) is commonly used to define elements, attributes, and data types.

Example:

```
<types>
  <schema xmlns="http://www.w3.org/2001/XMLSchema">
    <element name="GetDataRequest">
      <complexType>
        <sequence>
          <element name="id" type="int"/>
        </sequence>
      </complexType>
    </element>
  </schema>
</types>
```

2. MESSAGE

The Message component defines the information exchanged between the client and the service.

Example:

```
<message name="GetDataInput">
  <part name="request"
    element="GetDataRequest"/>
</message>

<message name="GetDataOutput">
  <part name="response"
    element="GetDataResponse"/>
</message>
```

3. PORTTYPE

PortType defines the operations offered by the web service. It specifies which messages are used as input and output.

Example:

```
<portType name="DataServicePortType">

  <operation name="GetData">
    <input message="GetDataInput"/>
    <output message="GetDataOutput"/>
  </operation>

  <operation name="UpdateData">
    <input message="UpdateDataInput"/>
    <output message="UpdateDataOutput"/>
  </operation>

</portType>
```

4. BINDING

Binding specifies how the service operations are transmitted over a communication protocol. SOAP over HTTP is commonly used.

Example:

```
<binding name="DataServiceBinding"
  type="DataServicePortType">

  <soap:binding
    style="document"
    transport="http://schemas.xmlsoap.org/soap/http"/>

</binding>
```

5. SERVICE

The Service component provides the actual network endpoint where the web service can be accessed.

Example:

```
<service name="DataService">

  <port name="DataServicePort"
        binding="DataServiceBinding">

    <soap:address
      location="http://example.com/DataService"/>

  </port>

</service>
```

COMPLETE WSDL STRUCTURE

```
<definitions>

  <types>
    <!-- XML Schema definitions -->
  </types>

  <message name="GetDataInput">
    <!-- Input message -->
  </message>

  <message name="GetDataOutput">
    <!-- Output message -->
  </message>

  <message name="UpdateDataInput">
    <!-- Input message -->
  </message>

  <message name="UpdateDataOutput">
    <!-- Output message -->
  </message>

  <portType name="DataServicePortType">

    <operation name="GetData">
      <input message="GetDataInput"/>
      <output message="GetDataOutput"/>
    </operation>
  </portType>
</definitions>
```

```

</operation>

<operation name="UpdateData">
  <input message="UpdateDataInput"/>
  <output message="UpdateDataOutput"/>
</operation>

</portType>

<binding name="DataServiceBinding"
  type="DataServicePortType">

  <soap:binding
    style="document"
    transport="http://schemas.xmlsoap.org/soap/http"/>

</binding>

<service name="DataService">

  <port name="DataServicePort"
    binding="DataServiceBinding">

    <soap:address
      location="http://example.com/DataService"/>

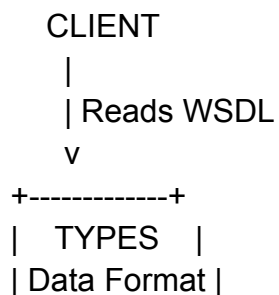
  </port>

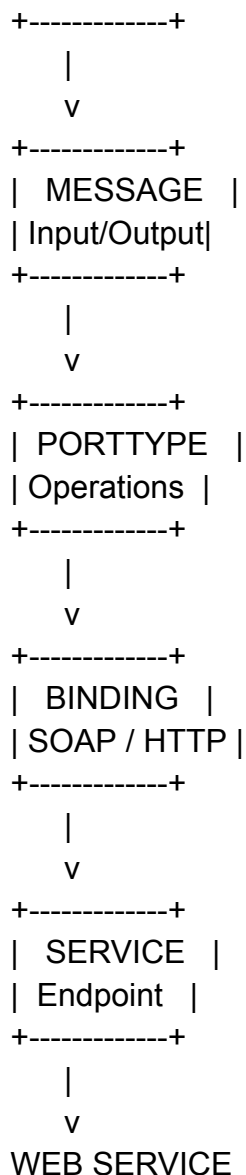
</service>

</definitions>

```

HOW WSDL DEFINES A SERVICE





EXAMPLE SERVICE OPERATIONS

Operation	Input	Output	Purpose
GetData	Data ID	Data details	Retrieves data
UpdateData	Data ID + new data	Status	Updates existing data
DeleteData	Data ID	Status	Deletes data
AddData	Data details	Data ID	Creates new data

IMPORTANCE OF WSDL IN SERVICE INTEGRATION

- WSDL provides a standard description of a web service.
- External applications can understand the service without knowing its internal implementation.
- It specifies exactly what operations are available and what data must be supplied.

- It allows applications developed in different programming languages to communicate.
- Development tools can use WSDL to automatically generate client-side code.
- WSDL reduces integration errors by clearly defining the service contract.

CONCLUSION

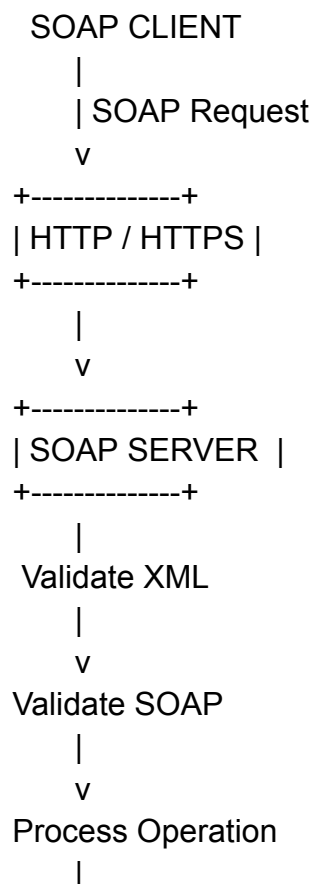
WSDL is important because it acts as a contract between a web-service provider and its clients. By defining types, messages, operations, bindings, and endpoints, WSDL makes service integration standardized, predictable, and interoperable.

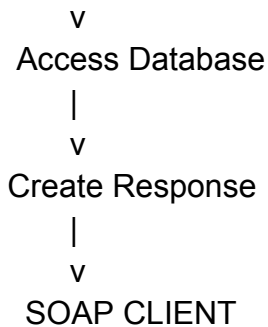
3. DEBUGGING SOAP COMMUNICATION PROBLEMS

INTRODUCTION

SOAP communication problems may occur when a request is incorrectly formatted, namespaces do not match, the endpoint is unavailable, or the server cannot process the requested operation. A systematic debugging process is required to identify the problem and restore reliable communication.

SOAP COMMUNICATION PROCESS





COMMON SOAP PROBLEMS

Problem	Possible Cause	Effect
Formatting error	Invalid XML or missing elements	Request rejected
Namespace error	Incorrect namespace URI	Operation not recognized
Endpoint error	Wrong URL or port	Server cannot be reached
SOAPAction error	Incorrect operation name	Wrong operation executed/rejected
Data type error	Wrong parameter type	Processing failure
Authentication error	Invalid credentials/token	Request rejected
Server error	Application/database failure	Incorrect or delayed response
Timeout	Network/server overload	Response delayed or lost

1. MESSAGE FORMATTING ERRORS

SOAP messages must follow the correct XML structure. Missing Envelope, incorrect Body structure, invalid XML syntax, or incorrect element names can cause the server to reject the message.

Incorrect example:

```

<soap:Envelope>
  <soap:Body>
    <GetData>
      <ID>101
    </GetData>
  </soap:Body>
</soap:Envelope>

```

The ID element is not properly closed, so the XML is invalid.

Correct example:

```

<soap:Envelope>

```

```
<soap:Body>
  <GetData>
    <ID>101</ID>
  </GetData>
</soap:Body>
</soap:Envelope>
```

2. NAMESPACE ERRORS

Namespaces identify XML elements and prevent conflicts between different XML vocabularies. If the client uses an incorrect namespace, the server may not recognize the SOAP operation.

Example:

Correct:

```
xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
```

Incorrect:

```
xmlns:soap="http://example.com/wrongnamespace"
```

The namespace must match the SOAP version and service definition expected by the server.

3. ENDPOINT PROBLEMS

The endpoint is the network address where the SOAP service is available. A wrong URL, port, path, or unavailable server can prevent communication.

Example:

Correct endpoint:

```
http://example.com/DataService
```

Possible incorrect endpoint:

```
http://example.com/DataService2
```

Endpoint checks include:

- Verify the service URL.
- Check whether the server is running.
- Check the port number.

- Check network connectivity.
- Verify firewall and proxy settings.

4. SOAPACTION AND OPERATION ERRORS

SOAPAction may identify the operation that the client wants to execute. If the SOAPAction or operation name is incorrect, the server may reject the request or execute the wrong operation.

Example:

SOAPAction: "GetData"

If the server expects:

SOAPAction: "UpdateData"

the request may not be processed correctly.

5. DATA TYPE AND PARAMETER ERRORS

The values supplied in the SOAP request must match the data types defined by the WSDL and XML Schema.

Example:

Expected:

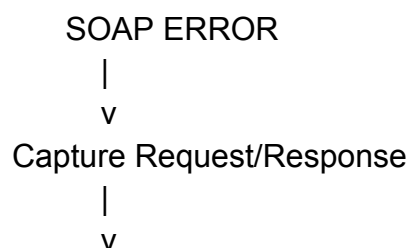
<PatientID>101</PatientID>

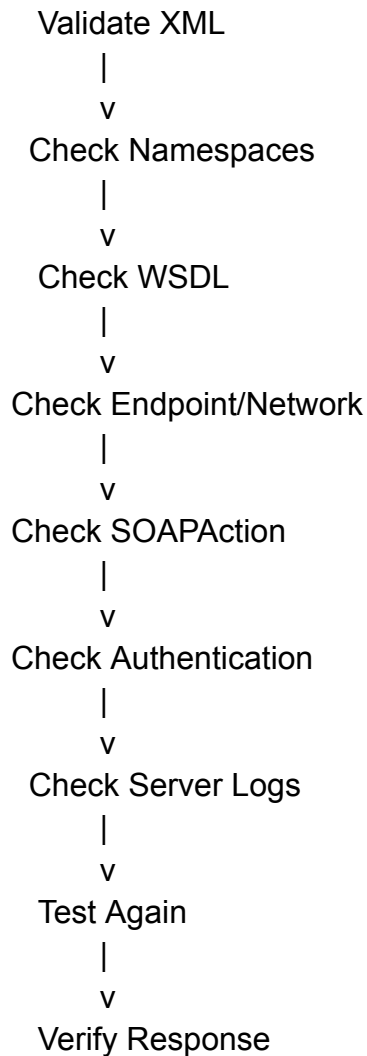
Incorrect:

<PatientID>ABC</PatientID>

if PatientID is defined as an integer.

DEBUGGING APPROACH





STEP-BY-STEP DEBUGGING

Step 1: Capture the SOAP request and response.

- Use a SOAP testing tool or network monitoring tool.
- Check the complete XML message.
- Compare the request with the expected structure.

Step 2: Validate the XML.

- Check whether the XML is well formed.
- Check opening and closing tags.
- Verify correct nesting of elements.
- Validate the message against the XML Schema.

Step 3: Verify namespaces.

- Check the SOAP namespace.
- Check service-specific namespaces.
- Compare namespaces with the WSDL.

- Correct any mismatched namespace.

Step 4: Verify the WSDL.

- Check operation names.
- Check input and output messages.
- Check data types.
- Check binding and service endpoint.

Step 5: Check network and endpoint.

- Verify the service URL.
- Check server availability.
- Verify HTTP/HTTPS configuration.
- Check firewall, proxy, and port settings.

Step 6: Check server-side processing.

- Examine server logs.
- Check authentication and authorization.
- Check database connectivity.
- Check application errors.

SOAP FAULT AND ERROR HANDLING

SOAP Fault is a standard mechanism used to communicate errors between a SOAP client and server.

SOAP Fault structure:

```
<soap:Fault>

  <faultcode>soap:Client</faultcode>

  <faultstring>
    Invalid Patient ID
  </faultstring>

  <detail>
    <errorCode>P001</errorCode>
  </detail>

</soap:Fault>
```

SOAP FAULT COMPONENTS

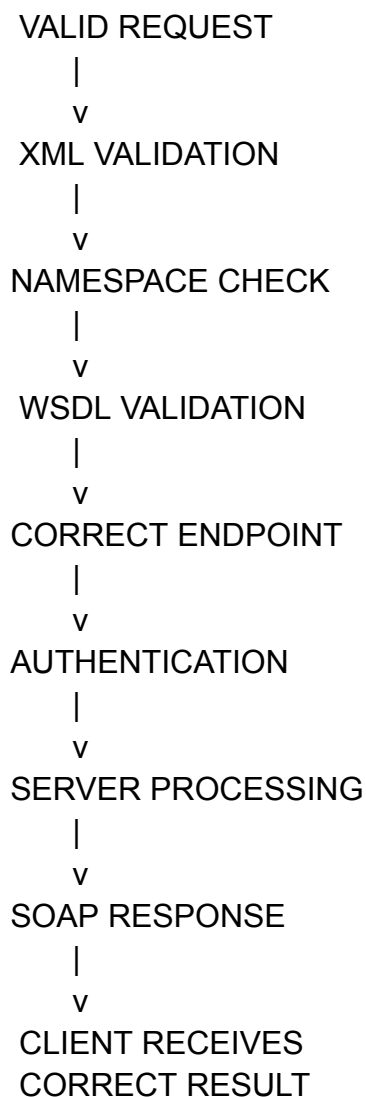
Component	Purpose
-----------	---------

|-----|-----|
| faultcode | Identifies the type/category of error |
| faultstring | Provides a human-readable error message |
| faultactor | Identifies the part of the system that caused the fault |
| detail | Provides application-specific error information |

COMMON FAULT TYPES

- Client Fault: The request contains invalid information.
- Server Fault: The server encountered an internal problem.
- Authentication Fault: User credentials or security information is invalid.
- Processing Fault: The server could not complete the requested operation.

RELIABILITY AFTER CORRECTION



HOW THE CORRECTED SYSTEM ENSURES RELIABLE COMMUNICATION

- XML validation prevents malformed messages from reaching the service.
- Correct namespaces ensure that SOAP operations are properly identified.
- WSDL validation ensures that requests contain the correct operations and data.
- Correct endpoint configuration ensures that requests reach the intended server.
- SOAP Faults provide standardized error information.
- Authentication and security checks prevent unauthorized requests.
- Server logs help administrators identify and fix recurring problems.
- Proper error handling allows clients to respond appropriately to failures.

CONCLUSION

Debugging a SOAP system requires checking the message structure, XML validity, namespaces, WSDL definitions, endpoint configuration, SOAPAction, authentication, and server-side processing. Using XML validation, WSDL verification, logging, monitoring, and SOAP Fault handling helps identify errors and ensures reliable, accurate, and secure communication between SOAP clients and servers.